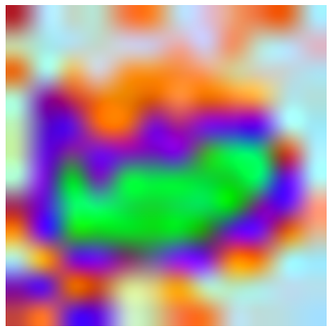
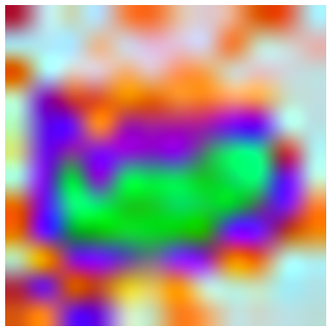
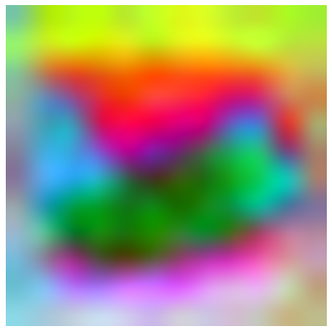
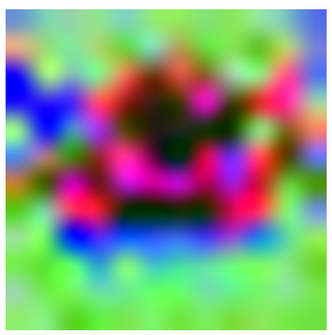
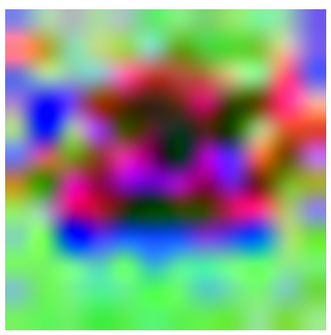
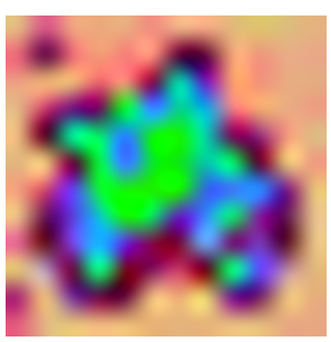
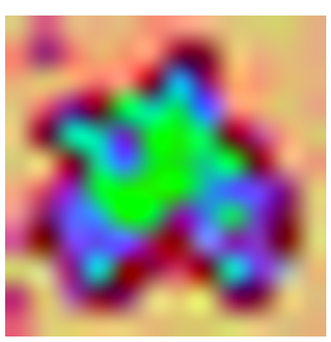
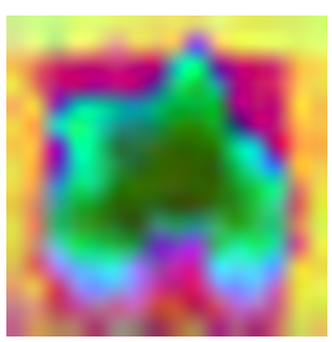
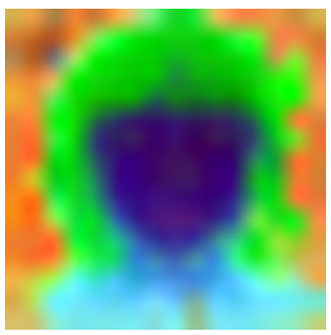
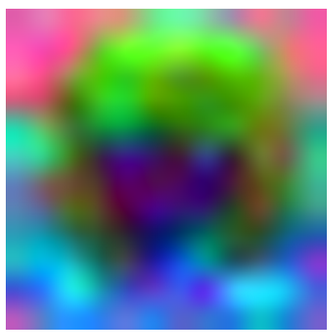
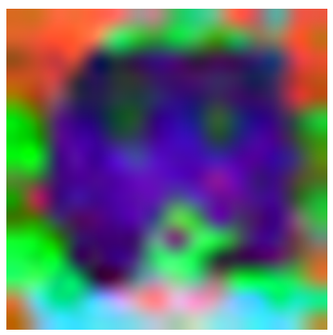
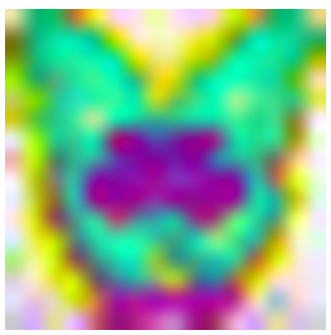
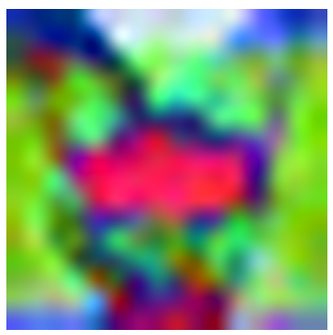
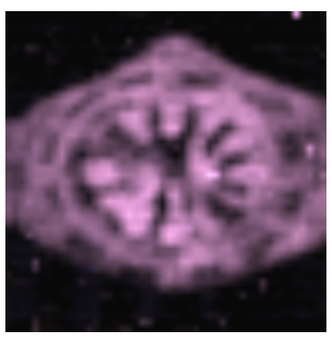
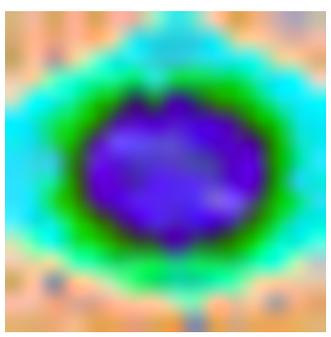
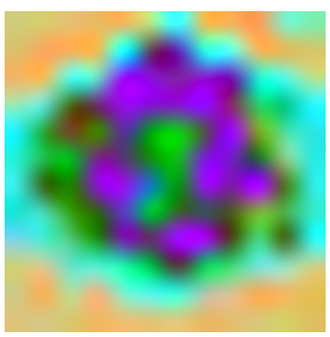
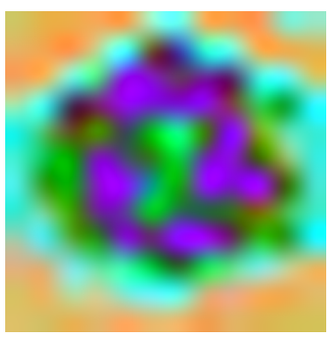
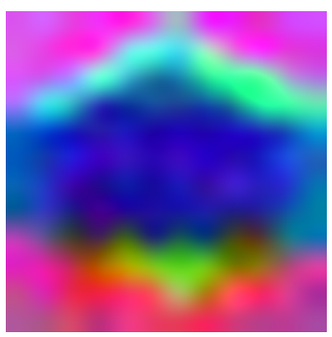
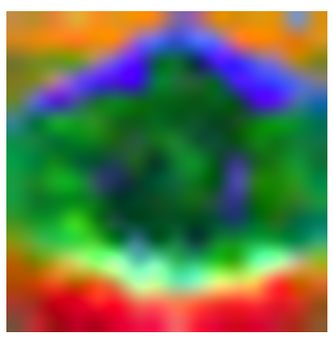
Input & Paired Reference | **DINO (ANN Teacher)** | **Ours: PureSpikeFormer Students (Event-Only Binary-Spike SNN, Ablation → Final)**

	(a) Paired RGB (reference only)	(b) Input Events ON / OFF (48×48)	**(c) DINOv2 Teacher ANN · 16×16 tokens PC1/2/3 = 24/13/10 %**	(d) Ours (Baseline) 12×12 · kNN 45.9% PC1/2/3 = 50/17/5 %	(e) Ours (+ Warm-Start) 12×12 · kNN 50.7% PC1/2/3 = 48/17/5 %	(f) Ours (Stage 2) 12×12 · kNN 61.4% PC1/2/3 = 15/8/6 %	**(g) Ours Final (Stage 3) 16×16 · kNN 63.7% PC1/2/3 = 18/8/6 %**
airplane							
automobile							
motorbike							
face							
butterfly							
watch							

DINO vs. Ours Guide: (c) DINOv2 Teacher = continuous floating-point DINOv2 (ViT-S/14) teacher (used only during training; not deployed at inference).
(d)–(g) Ours (PureSpikeFormer SNN) = our event-only pure binary-spike models: (d) Baseline, (e) + VP Warm-Start, (f) Stage 2 (+ dense patch affinity), and (g) Ours Final Stage 3 (16×16 + masked completion).